Reference Information Section

Engine Ratings

i07853902

Engine Rating Conditions

SMCS Code: 1000

It is important to know the use of the engine so that the rating will match the operating profile. The correct rating selection is also important so that the perception of price and value is realized by the customer.

In selecting a rating for a specific application, the most important consideration is the time that is spent at full throttle. The rating definitions identify the percent of time at full throttle. The definitions also identify the corresponding times below rated rpm.

Note: For an exact determination of the appropriate rating, follow the OEM specifications or consult your Caterpillar dealer.

The C3.6 industrial engine can have two ratings.

For C3.6 industrial engine with power rating from 56 to 90 kW (75 to 120.7 hp) the rating is C.

For C3.6 industrial engine with power ratings of 95 kW (127.4 hp) and 100 kW (134 hp), the rating is D.

For C2.8 industrial engine, the rating is C.

C – Intermittent service along with maximum power and/or speed are cyclic. The engine at full load should not exceed 50 percent of the duty cycle.

D – For service when the rated power is required for periodic overloads. The maximum horsepower and the rpm of the engine can be utilized continuously for a maximum of 30 uninterrupted minutes. This run is followed by one hour of operation at rating C. The engine should be run at full load. The engine should not exceed 10 percent of the duty cycle.

NOTICE
Operating engines above the rating definitions can result in shorter service life before overhaul.

i04650089

Engine Rating Definitions

SMCS Code: 1000

All engine ratings are in compliance with the following standard ambient air conditions of “ISO14396:2002”:

- 100 kPa (29.3 Inches Hg)
- 30 percent relative humidity
- A temperature of 25 °C (77 °F)

The engine ratings are based on the following fuel specifications:

- The Low Heat Value (LHV) of the fuel at 29 °C (84.2 °F) at 42780 kJ/kg (18390 Btu/lb)
- Specific gravity of .833 - 837 at 15 °C (60 °F)

The engine ratings are gross output ratings.

Gross Output Ratings – The total output capability of the engine that is equipped with standard accessories.

Standard accessories include the following components:

- Oil pumps
- Fuel pumps
- Water pumps

Subtract the power that is required to drive auxiliary components from the gross output. This process will produce the net power that is available for the external load (flywheel).

Customer Service

i07191578

Customer Assistance

SMCS Code: 1000; 4450

USA and Canada

When a problem arises concerning the operation or the service of an engine, the problem will normally be managed by the dealer in your area.

Outside of the USA and of Canada

If a problem arises outside the USA and outside Canada, and if the problem cannot be resolved at the dealer level, consult the appropriate Caterpillar office.

Latin America, Mexico, Carribean
Caterpillar Americas Co.
701 Waterford Way, Suite 200
Miami, FL 33126-4670
USA
Phone: 305-476-6800
Fax: 305-476-6801

Europe, Africa, and Middle East
Caterpillar Overseas S.A.
76 Route de Frontenex
P.O. Box 6000
CH-1211 Geneva 6
Switzerland
Phone: 22-849-4444
Fax: 22-849-4544

Far East
Caterpillar Asia Pte. Ltd.
7 Tractor Road
Jurong, Singapore 627968
Republic of Singapore
Phone: 65-662-8333
Fax: 65-662-8302

China
Caterpillar China Ltd.
37/F., The Lee Gardens
33 Hysan Avenue
Causeway Bay
G.P.O. Box 3069
Hong Kong
Phone: 852-2848-0333
Fax: 852-2848-0440

Australia and New Zealand
Caterpillar of Australia Ltd.
1 Caterpillar Drive
Private Mail Bag 4
Tullamarine, Victoria 3043
Australia
Phone: 03-9953-9333
Fax: 03-9335-3366

i03655027

Ordering Replacement Parts

SMCS Code: 4450; 7567

⚠ WARNING

When replacement parts are required for this product Caterpillar recommends using Caterpillar replacement parts or parts with equivalent specifications including, but not limited to, physical dimensions, type, strength and material.

Failure to heed this warning can lead to premature failures, product damage, personal injury or death.

Quality Caterpillar replacement parts are available from Caterpillar dealers throughout the world. Caterpillar dealers' parts inventories are up-to-date. The parts stocks include all of the parts that are normally needed to protect your Caterpillar engine investment.

When you order parts, please specify the following information:

- When a Caterpillar engine requires maintenance and/or repair, provide the dealer with all the information that is stamped on the Information Plate. Refer to this Operation and Maintenance Manual, “Plate Locations and Film Locations”.
- Part number
- Part name
- Quantity

If there is a question concerning the part number, please provide your dealer with a complete description of the needed item.

Discuss the problem with the dealer. Inform the dealer about the conditions of the problem and the nature of the problem. Inform the dealer about when the problem occurs. This will help the dealer in troubleshooting the problem and solving the problem faster.

Reference Materials

i08530562

Maintenance Records

SMCS Code: 1000; 4450

Caterpillar recommends the retention of accurate maintenance records. Accurate maintenance records can be used for the following purposes:

- Determine operating costs.
- Establish maintenance schedules for other engines that are operated in the same environment.
- Show compliance with the required maintenance practices and maintenance intervals.

Maintenance records can be used for various other business decisions that are related to engine maintenance.

Maintenance records are a key element of a maintenance program that is managed. Accurate maintenance records can help your Cat dealer to fine-tune the recommended maintenance intervals to meet the specific operating situation. These recommendations should result in a lower engine operating cost.

Records should be kept for the following items:

Fuel Consumption – A record of fuel consumption is essential to determine when the load sensitive components should be inspected or repaired. Fuel consumption assists in determining overhaul intervals.

Service Hours – A record of service hours is essential to determine when the speed sensitive components should be inspected or repaired.

Documents – These items should be easy to obtain, and these items should be kept in the engine history file. All documents should show this information: date, service hours, fuel consumption, unit number, and engine serial number. The following types of documents should be kept as proof of maintenance or repair for warranty:

Keep the following types of documents as proof of maintenance for warranty. Also, keep these types of documents as proof of repair for warranty:

- Dealer work orders and itemized bills
- Owner repair costs
- Owner receipts
- Maintenance log

i08254037

Maintenance Log

SMCS Code: 1000; 4450

Table 29

Engine Model		**Customer Identifier**		
Serial Number		**Arrangement Number**		
Service Hours	**Quantity Of Fuel**	**Service Item**	**Date**	**Authorization**

i07853896

Reference Material

SMCS Code: 1000; 4450

The following literature can be obtained through any Caterpillar dealer.

Lubricants

- Operation and Maintenance Manual, SEBU5898, “Cold-Weather Recommendations”
- Operation and Maintenance Manual, SEBU6251, “Caterpillar Commercial Diesel Engine Fluids Recommendations”
- Special Publication, PEHP6001, “How To Take A Good Oil Sample”

Coolants

- Special Publication, PEHP4036, “Data Sheet - Extended Life Coolant”
- Special Publication, PEHP7057, “Data Sheet - S·O·S Coolant Analysis”
- Special Publication, SEBD0518, “Know Your Cooling System”
- Label, PEEP5027, “Extended Life Coolant/ Antifreeze”

Miscellaneous

- Service Manual, REG1139F, “Service Manual Contents Microfiche”
- Service Manual, UENR7548, “ C3.6 and C2.8 Industrial Engines”
- Troubleshooting, M0107940, “ C3.6 Engines”
- Troubleshooting, M0112481, “ C2.8 Engines”
- Systems Operation, Testing, and Adjusting, M0107832, “ C3.6 and C2.8 Industrial Engines”
- Specifications, M0107642, “ C3.6 Industrial Engines.”
- Specifications, M0109805, “ C2.8 Industrial Engines.”
- Disassembly and Assembly, M0108719, “ C3.6 and C2.8 Industrial Engines”
- Specifications, SENR3130, “Torque Specifications”
- Special Publication, PECP9067, “One Safe Source” English language for use in NACD
- Special Publication, SEBF8029, “Index to Guidelines for Reusable Parts and Salvage Operations”
- Special Publication, SEBF8062, “Procedure to Inspect and Clean Air Filters”
- Special Instruction, SEHS9031, “Storage Procedure for Caterpillar Products”
- Special Publication, NEHS0526, “Service Technician Application Guide”
- Special Publication, SEBU6251, “Caterpillar Commercial Diesel Engine Fluids Recommendations”
- Special Instruction, SEHS7633, “Battery Test Procedure”

Index

A

Additional Messages 16
After Starting Engine 71
After Stopping Engine 88
Aftercooler Core - Clean/Test (Air-To-Air Aftercooler) 104
Aftercooler Core - Inspect 104
Aftertreatment Operation 76
Alarms and Shutoffs 53
  Alarms 53
  Shutoffs 53
  Testing 54
Alternator - Inspect 104
Alternator and Fan Belts - Replace 104

B

Battery - Replace 104
Battery Disconnect Switch (If Equipped) 54
Battery Electrolyte Level - Check 105
Battery or Battery Cable - Disconnect 105
Before Starting Engine 27, 69
Belt - Inspect 106
Belt Tensioner - Check 107
Burn Prevention 21
  Batteries 22
  Coolant 22
  Diesel Fuel 22
  Engine and Aftertreatment System 22
  Induction System 22
  Oils 22

C

Cold Weather Operation 83
  Hints for Cold-Weather Operation 83
  Idling the Engine 84
  Recommendations for Coolant Warm Up 84
  Recommendations for the Coolant 84
  Viscosity of the Engine Lubrication Oil 84
Cold Weather Starting 69
Coolant (DEAC) - Change 107
  Cooling Systems with Heavy Deposits or Plugging 111
  Drain 108
  Fill 111
  Flush 110
Coolant (ELC) - Change 112
  Drain 112
  Fill 114
  Flush 114
Coolant Extender (ELC) - Add 115
Coolant Level - Check 116
  Open Power Unit (OPU) Radiators 116
Coolant Sample (Level 1) - Obtain 117
Coolant Sample (Level 2) - Obtain 118
Cooling System Supplemental Coolant Additive (SCA) - Test/Add 118
  Add the SCA, If Necessary 119
  S·O·S Coolant Analysis 118
  Test for SCA Concentration 118
Crushing Prevention and Cutting Prevention 25
Customer Assistance 152
  Outside of the USA and of Canada 152
  USA and Canada 152
Customer Service 152

D

DEF Filler Screen (Emission Related Component) - Clean 119
DEF Manifold Filters (Emission Related Component) - Replace 120
Diagnostic Flash Code Retrieval 67
Diagnostic Lamp 67
Diesel Exhaust Fluid (Emission Related Component) - Fill 120
Diesel Exhaust Fluid Filter (Emission Related Component) - Replace 121
Diesel Exhaust Fluid in Cold Weather 86
Diesel Exhaust Fluid Injector (Emission Related Component) - Replace 122
Diesel Exhaust Fluid Tank - Flush 123
  Flushing Procedure 123
  Required Tools 123
Driven Equipment - Check 125

E

Electrical System 28
  Grounding Practices 28
Emergency Stopping 88
Emissions Certification Film 47
Emissions Warranty Information 150
Engaging the Driven Equipment 74
Engine - Clean 125
  Aftertreatment 125
Engine Air Cleaner Element - Replace 125

Servicing the Air Cleaner Elements 125
Engine Air Cleaner Service Indicator - Inspect 127
Test the Service Indicator 127
Type 1 Service Indicator 127
Type 2 Service Indicator 127
Engine Air Precleaner - Check/Clean 128
Engine Diagnostics 67
Engine Electronics 29
Engine Mounts - Inspect 128
Engine Oil and Filter - Change 130
Drain the Engine Lubricating Oil From a Standard Oil Pan 130
Draining the Engine Lubricating Oil From an Oil Pan Equipped With a Balancer 131
Fill the Oil Pan 133
Oil and Filter Change Intervals 130
Replace the Oil Filter Element 133
Engine Oil Level - Check 128
After Operating the Engine 129
Before Operating the Engine 128
Engine Oil Sample - Obtain 129
Obtain the Sample and the Analysis 129
Engine Operation 73
C2.8 Industrial Engine 74
C3.6 Industrial Engine 74
Carbon Dioxide (CO_2) Emissions Statement 74
Engine Operation and the Aftertreatment System 73
Engine Operation with Active Diagnostic Codes 67
Engine Operation with Intermittent Diagnostic Codes 68
Engine Rating Conditions 151
Engine Rating Definitions 151
Engine Ratings 151
Engine Starting 27, 69
Engine Stopping 28, 88

F

Fan Clearance - Check 134
C2.8 Open Power Unit (OPU) 134
C3.6TA Open Power Unit (OPU) 134
Fault Logging 67
Features and Controls 53
Fire Prevention and Explosion Prevention 23
Ether 24
Fire Extinguisher 24
Lines, Tubes, and Hoses 25
Fluid Recommendations 94
Coolant 99
Diesel Exhaust Fluid 97
Engine Lubrication Oil 94
Fuel 97
Lubricating Grease 96
S·O·S Coolant Analysis 100
Foreword 4
California Proposition 65 Warning 4
Literature Information 4
Maintenance 4
Maintenance Intervals 5
Operation 4
Overhaul 5
Safety 4
Fuel and the Effect from Cold Weather 85
Fuel Conservation Practices 74
Fuel Related Components in Cold Weather 86
Fuel Filters 86
Fuel Heaters 86
Fuel Tanks 86
Fuel System - Prime 135
Electric Fuel Priming Pump 135
Mechanical Fuel Priming Pump 135
Fuel System Primary Filter (Water Separator) Element - Replace 136
Fuel System Primary Filter (Water Separator) Element for Electric Fuel Priming Pump .. 136
Fuel System Primary Filter (Water Separator) Element for Manual Fuel Priming Pump .. 137
Fuel System Primary Filter/Water Separator - Drain 139
Drain Procedure 139
Fuel System Secondary Filter - Replace 139
Install the Element 140
Remove the Element 140
Fuel Tank Water and Sediment - Drain 141
Drain the Water and the Sediment 141
Fuel Storage Tanks 141
Fuel Tank 141

G

Gauges and Indicators 54
Aftertreatment Lamps and gauges 56
Indicator Lamps 55
General Hazard Information 17
Containing Fluid Spillage 19
Diesel Exhaust Fluid 21
Dispose of Waste Properly 21
Fluid Penetration 19
Inhalation 20
Pressurized Air and Water 18

Static Electricity Hazard when Fueling with Ultra-low Sulfur Diesel Fuel 19
General Information 31

H

High Pressure Fuel Lines 25
Hoses and Clamps - Inspect/Replace 141
Replace the Hoses and the Clamps 142

I

Important Safety Information 2

L

Lifting and Storage 49

M

Maintenance Interval Schedule 102
Daily 102
Every 10 000 Service Hours 103
Every 1000 Service Hours 102
Every 12 000 Service Hours or 6 Years 103
Every 2000 Service Hours 102
Every 250 Service Hours 102
Every 3000 Service Hours 102
Every 3000 Service Hours or 3 Years 103
Every 4000 Service Hours 103
Every 4500 Service Hours 103
Every 50 Service Hours 102
Every 500 Service Hours or 1 Year 102
Every 6000 Service Hours 103
Every 6000 Service Hours or 3 Years 103
Every Week 102
Every Year 102
Overhaul 103
When Required 102
Maintenance Log 154
Maintenance Recommendations 90
Maintenance Records 153
Maintenance Section 90
Model View Illustrations 31, 36, 39
C2.8 Industrial Engine Views 33
C2.8 OPU 42
C3.6 Industrial Engine Views 31
C3.6 Industrial Engine Views For Engines Equipped With A Balancer 36
C3.6 Open Power Unit (OPU) 39
Loose or Off Engine Components 35, 38
Monitoring System (Engine Indicators and Aftertreatment Indicators) 56
Engine Indicator Lamps 56
Mounting and Dismounting 25

O

Operation Section 49
Ordering Replacement Parts 152
Overhaul Considerations 142
Oil Consumption as an Overhaul Indicator 143
Overhaul Recommendation 143

P

Plate Locations and Film Locations 47
Serial Number Plate 47
Product Description ((Engine and Aftertreatment)) 44
Aftermarket Products and Caterpillar Engines 46
Aftertreatment System 46
C2.8 Industrial Engine 44
C3.6 Industrial Engine 44
Electronic Engine Features 45
Engine Cooling and Lubrication 45
Engine Diagnostics 45
Engine Service Life 46
Engine Specifications 44
Product Identification Information 47
Product Information Section 31
Product Lifting 49
C2.8 Industrial Engine 51
C2.8 Open Power Unit (OPU) 51
C3.6 Industrial Engine 49
C3.6 Industrial Engine Equipped With A Balancer 50
C3.6 Open Power Unit (OPU) 50
Product Storage 51
Aftertreatment 52

R

Radiator - Clean 145
Radiator Mounts - Inspect 146
Radiator Restrictions 85
Reference Information 48
Record for Reference 48
Reference Information Section 151
Reference Material 155
Coolants 155

Lubricants 155
Miscellaneous 155
Reference Materials 153
Refill Capacities 93
Cooling System 94
DEF System 94
Lubrication System for Engines Equipped With a Balancer Oil Pan 93
Lubrication System for Engines Equipped With a Standard Oil Pan 93

S

Safety Messages 6, 10, 13
C2.8 Industrial Engine 8
C3.6 Industrial Engine 7
C3.6 Industrial Engine With A Balancer 11
C3.6 Open Power Unit (OPU) 14
Ether Warning 9, 12, 16
Hand (High Pressure) 9
Hand (High Pressure) 2 12
Hand (High Pressure) 3 15
Hot Fluid Under Pressure 2 15
Universal Warning 8
Universal Warning 1 ..... 11, 14
Safety Section 6
Selective Catalytic Reduction Warning System 76
Warning Indicators 76
Warning Levels 76
Warning Strategy 76
Self-Diagnostics 67
Sensors and Electrical Components 57
C2.8 Industrial Engine 63
C3.6 Industrial Engines 58
Off Engine Sensors and Electrical Components 62
Severe Service Application 92
Improper Maintenance Procedures (Maintenance Procedures Which May Contribute to a Severe Service Application) 92
Severe Environmental Factors 92
Severe Operating Conditions 92
Starting Motor - Inspect 146
Starting the Engine 70
Starting the Engine 70
Starting with Jump Start Cables (Do Not Use This Procedure in Hazardous Locations that have Explosive Atmospheres) 70
Stopping the Engine 88
Delayed Engine Shutdown (if Equipped) 88
System Pressure Release 90
Coolant System 90
Engine Oil 90
Fuel System 90

T

Table of Contents 3
Turbocharger - Inspect 146

W

Walk-Around Inspection 147
High-Pressure Fuel Lines 148
Inspect the Engine for Leaks and for Loose Connections 147
Warranty Information 150
Warranty Section 150
Water Pump - Inspect 148
Welding on Engines with Electronic Controls 90

Product and Dealer Information

Note: For product identification plate locations, see the section “Product Identification Information” in the Operation and Maintenance Manual.

Delivery Date: _____

Product Information

Model: _____

Product Identification Number: _____

Engine Serial Number: _____

Transmission Serial Number: _____

Generator Serial Number: _____

Attachment Serial Numbers: _____

Attachment Information: _____

Customer Equipment Number: _____

Dealer Equipment Number: _____

Dealer Information

Name: _____ Branch: _____

Address: _____

_____

_____

	Dealer Contact	Phone Number	Hours
Sales:	_____	_____	_____
Parts:	_____	_____	_____
Service:	_____	_____	_____

M0108394
©2022 Caterpillar
All Rights Reserved

CAT, CATERPILLAR, LET'S DO THE WORK, their respective logos, "Caterpillar Corporate Yellow", the "Power Edge" and Cat "Modern Hex" trade dress as well as corporate and product identity used herein, are trademarks of Caterpillar and may not be used without permission.